Abstract – Online Rental House Management System

The project is designed to simplify the process of renting houses by allowing users to search available rental properties, view house details, and manage bookings through a web application without manual paperwork. The system aims to automate the traditional house rental process, reducing time consumption and improving accuracy in property and tenant management. It helps administrators manage house details, tenant information, rental requests, payments, and property availability in an organized manner. The project also enhances user convenience by providing easy access to rental house information, pricing, location details, and booking services through the internet.

Modules:

1. Administrator

- Manages the overall system
- Adds, updates, and removes property details
- Monitors rental requests and tenant information
- Maintains property availability and system operations

2. Tenants / Users

- Register and log into the system
- Browse available rental houses and rent payment
- View property details, pricing, and location
- Apply for house rentals and manage bookings

3. Property Owners

- Maintain house and property information
- Track rental status and availability
- Handle tenant requests and approvals
- Manage rental payments and agreements

Main Functionalities

- User dashboard for registration and secure login
- Property listing with images and pricing
- House availability checking
- Online rental booking and request management
- Rental payment management
- Admin dashboard for managing properties and tenants
- Property owner dashboard for managing houses and bookings
- Responsive frontend interface
- REST API integration between frontend and backend
- Secure authentication

The endpoints are commonly used in rental house management applications for handling authentication, property management, rental bookings, tenant records, and payment operations. These endpoints form the core communication layer of the Online Rental House Management System and contribute significantly to providing a responsive, efficient, and user-friendly property rental platform.

API Endpoints Included

- POST /api/register/ – User registration
- POST /api/login/ – User authentication
- GET /api/properties/ – Retrieve available rental houses
- GET /api/properties/{id}/ – View property details
- POST /api/bookings/ – Create rental booking request
- GET /api/bookings/ – Retrieve booking history
- PUT /api/bookings/{id}/ – Update booking details
- DELETE /api/bookings/{id}/ – Cancel booking request
- GET /api/users/ – Retrieve tenant details
- POST /api/payments/ – Handle rental payment processing
- GET /api/admin/dashboard/ – Admin management dashboard

Technologies Used

- Django
- React
- PostgreSQL
- HTML
- CSS
- JavaScript
- Bootstrap

The primary objective of this project is to develop a scalable and modern rental house management solution that fulfills the growing need for digital property rental services. The project demonstrates the practical implementation of full-stack web development concepts, authentication systems, REST API integration, database management, and software engineering principles in a real-world application environment within Software Engineering.